Georgy Savva

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Research Interests

World Models, Representation Learning, Multimodal Foundatinal Models, Agent Learning

Education

New York University — New York, United States
Master of Science in Computer Science, Courant Institute of Mathematical Sciences. GPA: 3.8 — *2024 – 2025*

Russian University of Cooperation — Moscow, Russia
Bachelor of Science in Computer Science — *2019 – 2021*

National Research Nuclear University MEPhI — Moscow, Russia
Bachelor of Science in Cybersecurity — *2014 – 2016*

Experience

Member of Technical Staff, Research — Jun 2026 – Present
Runway — *New York, United States*

- Foundational video world models for robotics

Research Assistant — Jan 2025 – Feb 2026
NYU Courant Institute of Mathematical Sciences, Advised by Prof. Saining Xie — *New York, United States*

- Developed the first multi-agent Minecraft World Model
- Investigated quality degradation in video world models by applying Diffusion Forcing and Self Forcing

Research Intern — Jun 2024 – Nov 2024
NYU Courant Institute of Mathematical Sciences, Advised by Prof. Lerrel Pinto — *New York, United States*

- Applied online reinforcement learning to fix the human demonstration morphology gap for dexterous robotic manipulation
- Trained a behavior cloning policy using transformers on third-person human demonstration data to solve dexterous robotic hand tasks
- Set up an auto-resettable reinforcement learning environment for online robot training using the spacemouse device

Senior Software Engineer — Oct 2023 – May 2024
Raylu — *New York, United States*

- Led backend development of an LLM chatbot for healthcare and deployed it to production, serving first paying customers
- Developed an LLM-powered SaaS workflow automation application using the open source workflow engine Activepieces, allowing the company to release a new product in under 3 months

Technical Co-Founder — Jul 2022 – Aug 2023
Scifind — *Los Angeles, United States*

- Launched a troubleshooting platform for bioscientists into production, gaining 4,000 MAU in the first 2 months
- Led the development of the product with a team of 2 engineers using TypeScript, Next.js, and Node.js

Senior Software Engineer — Jan 2021 – Jul 2022
IOTA — *Berlin, Germany*

- Developed a new network layer of the blockchain node using libp2p, decreasing the number of peering errors by 70%
- Set up automatic deployment via Ansible and GitHub Actions, accelerating the development speed by 25%
- Introduced Go guidelines to the project and tools to ensure them, reducing the frequency of bugs by 50%

Senior Software Engineer — Jan 2020 – Jan 2021
Elsa — *San Francisco, United States*

- Implemented App Store and Google Play in-app subscriptions as microservices using FastAPI, which improved the correctness of billing by 50%

- Integrated payment gateway Instamojo, increasing the number of app purchases in Asia threefold

Software Engineer Mar 2017 – May 2019
Edwin *San Francisco, United States*

- Participated in building the English tutor chatbot from a prototype stage to being the top 1 bot on Facebook
- Implemented a YAML-file-based framework for building dialog systems, which allowed the company to launch a new product on Google Assistant
- Designed a distributed queue using PostgreSQL to handle messages from 800,000 users
- Developed the user knowledge graph service using Neo4j, improving the performance of read queries threefold

Software Engineer Jun 2016 – Mar 2017
Snaappy *Moscow, Russia*

- Rewrote the messaging service in Go, which increased the backend performance fivefold
- Migrated the chat storage from PostgreSQL to MongoDB, allowing horizontal scalability for the data layer
- Implemented a Websocket service for real-time updates using Go, driving user engagement by 30%

Teaching Experience

Graduate Teaching Assistant Sep 2025 – Dec 2025
NYU Courant Institute of Mathematical Sciences. Prof. Saining Xie *New York, United States*

- Led weekly office hours for CSCI-GA.2271-001 Computer Vision. Website
- Helped students with their research projects and homework

Publications

1. *Solaris: Building a Multiplayer Video World Model in Minecraft*
   G. Savva*, O. Michel*, D. Lu*, P. Waiwitlikhit, T. Meehan, D. Mishra, S. Poddar, J. Lu, S. Xie.
   Under Review 2026. Website

2. *HuDOR: Bridging the Human to Robot Dexterity Gap through Object-Oriented Rewards*
   I. Guzey, Y. Dai, **G. Savva**, R. Bhirangi, L. Pinto.
   ICRA 2025. Website

Projects

Do Pre-Trained and Fine-Tuned World Models Generalize? Jan 2025 – May 2025
Investigated the generalization and finetuning of two SOTA Minecraft World Models on three data distributions. Website

AppSim: A Learned World Model for an App API Jan 2025 – May 2025
Used ChatGPT o3 in a zero-shot setting, achieving 74% accuracy, and compared its performance to a finetuned TinyLlama model. Website

Transformer-Based Diffusion for Game Generation Oct 2024 – Dec 2024
Trained a transformer diffusion model to simulate DOOM trajectories. It achieves a PSNR of 32.21 in the teacher-forcing setting, producing an indistinguishable quality from the ground truth. Website

Advantage Actor-Critic with Optuna Jul 2024 – Sep 2024
Trained a reinforcement learning agent with Optuna for the HalfCheetah env, achieving a 24% better performance than the best publicly available policy. Website

Open-Source Library Scany May 2020 – Present
Created an open-source library to map data from a database into Go objects. The library has 1,500 stars on GitHub and is used by thousands of companies. GitHub

Skills

Languages: Python, SQL, Go, TypeScript
Machine Learning: Transformers, Diffusion, U-Net, CNN, LSTM, Reinforcement Learning, Distributed Training, TPU
Libraries: PyTorch, JAX, Gym, Optuna, Hydra, TensorFlow, Numpy, Pandas, FastAPI, React, Node.js
Tools: Docker, Kubernetes, PostgreSQL, AWS, GraphQL, Hasura